

Piyush Singh

CEH | CRTP | [Linkedin](#)

Application Security Engineer | Penetration Testing

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- **Professional Summary:** Application Security Engineer with hands-on experience securing **web, API, mobile, and infrastructure environments** across consulting and product setups. Delivered **end-to-end penetration testing, vulnerability triage, and remediation guidance** for enterprise and critical infrastructure clients, including power plants. Strong focus on **OWASP Top 10, CI/CD security integration, DevSecOps enablement, and automation-driven testing**. Recognized through multiple **Hall of Fame acknowledgments** from global organizations for valid, impactful security findings.
- Application Security Testing (Web, API, Mobile) • OWASP Top 10, CVE, CWE, CVSS • Secure SDLC & DevSecOps Enablement • SAST, DAST, Dependency & Config Reviews • Vulnerability Triage, Validation & Reporting • Red Teaming Fundamentals & AD Exposure • Security Automation (Python, Bash) • Stakeholder & Developer Risk Communication.

ACHIEVEMENTS

- Bug Bounty / HOFs / Recognition: - **Nvidia, Ferrari, Bosch, Philips, Michelin, Inflectra, Redvillia**
- Reported validated security vulnerabilities to global organizations
- Core organizing member, Technovation Hackathon

EXPERIENCE

Confidential - Penetration Testing – ABU DHABI

02-2026 - Present

- Led penetration testing and Red Team engagements across infrastructure, web, mobile, and networks to identify exploitable vulnerabilities and assess security posture
- Developed exploits, testing methodologies, and conducted remediation validation to ensure effective vulnerability closure.
- Delivered risk-focused reports and communicated findings to both technical and business stakeholders for actionable remediation.
- Performed threat modeling, secure code reviews, and mentored junior team members to strengthen overall security maturity.

Ralfkairos - Application Security Engineer – SOUTH KOREA

07-2025– 01-2026

- Performed security assessments across web applications, APIs, internal networks, and mobile applications (**iOS and Android**) in pre-production and production-adjacent environments, identifying OWASP Top 10 issues, **authentication and authorization flaws**, IDOR, misconfigurations, and business logic vulnerabilities.
- Built and maintained repeatable testing environments and exploitation scenarios to reliably **validate complex vulnerabilities** across web, network, and mobile surfaces, ensuring accuracy of findings before reporting.
- Improved testing efficiency and signal quality by standardizing reconnaissance and exploitation workflows and evaluating automation-assisted testing approaches (for ex: **Xbow**) to reduce repetitive manual effort while preserving manual validation for high-impact findings.

Cybersrc - Associate Security Analyst - INDIA

01-2025–07-2025

- Performed **Vulnerability Assessment and Penetration Testing (VAPT)** across web applications, infrastructure, and internal networks for 30+ clients. Conducted onsite security assessments at 10+ locations, including **critical infrastructure environments** such as **power plants**.
- Supported compliance teams through **technical evidence validation**, scope clarification, and infrastructure risk interpretation.
- Managed **multiple security engagements** simultaneously, handling client communication, status updates, and delivery timelines.
- Developed custom scripts, proof-of-concept exploits, and automation tooling to validate emerging vulnerabilities to reduce manual effort.

SECURITY PROJECTS

- **Tracedrill:** Security automation platform focused on **streamlining reconnaissance, misconfiguration detection, and OWASP Top 10 testing**. Built automated workflows performing **40+ security checks** across target applications. Reduced manual reconnaissance and triage effort for testers. Designed for developers, bug bounty hunters, and security teams to accelerate early vulnerability discovery.
- **Recon-Automator:** Developed a CLI-based reconnaissance tool for automated vulnerability detection with a Web based Interface too, improving efficiency by 40%. Reduced manual reconnaissance efforts by 40%, enhancing team productivity. Leveraged Python for robust automation and improved tool scalability.
- **Google Dorking Query Generator:** OSINT automation tool simplifying complex Google dork creation. Enabled rapid discovery of exposed assets, misconfigured servers, and sensitive files during reconnaissance.

EDUCATION

Bachelor of Technology, Computer Science, Sharda University

Sept 2021 —2025

TECHNICAL SKILLS

- Security Testing: Web, API, Mobile (Android/iOS), Network, Wireless.
- Tooling: Burp Suite, Nmap, Wireshark, Metasploit, Nessus, SQLmap, BloodHound, Impacket, CrackMapExec, FFUF, Amass
- Automation: Python, Bash Web & Code: HTML, CSS, JavaScript, SQL
- Platforms: CI/CD pipelines, GitHub/GitLab, Linux, Windows AD

CERTIFICATIONS

- CEH - Certified Ethical Hacker
- CRTP - Certified Red Team Professional
- Practical Web Application Security (TCM Security),
- Windows and Linux Privilege Escalation for Beginners (TCM Security),
- Cybersecurity Fundamentals – IBM